Valerie Fong

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Education

University of California, Davis, Ph.D., Applied Mathematics — Expected Dec 2027
Dissertation: Temporal Computation through Dynamically Critical Networks. Advisor: Rishidev Chaudhuri.

University of California, Davis, M.S., Applied Mathematics — 2024

University of New Mexico, B.S., Statistics and Economics, minor in Mathematics — 2022
Phi Beta Kappa

Fellowships: NSF Graduate Research Fellowship; National GEM Consortium PhD Fellowship; UC Davis Hazel B. Jacoby Fellowship

Experience

iRhythm Technologies, Data Science Intern — Jun – Aug 2022

- Built and benchmarked demand-forecasting models (ARIMA, convolutional neural networks, empirical mode decomposition, seasonal-trend decomposition), comparing them across multiple error metrics to improve forecast accuracy.
- Applied k-means clustering and PCA to identify demand trends.

Sandia National Laboratories, Year-Round R&D Intern — 2019 – 2021

- Built and maintained a data visualization dashboard of safety, security, quality, and environmental metrics used to support leadership decisions.
- Supported cost-benefit analyses for a DOE Carbon Capture, Utilization and Storage program spanning multiple national laboratories, geological surveys, and universities.
- Earned Lean Six Sigma Green Belt certification. Held a DOE Q clearance.

Research

UC Davis, Graduate Student Researcher, Department of Mathematics — 2022 – Present

- Develop theory and simulations of recurrent neural networks near a dynamical phase transition, studying how criticality supports long-timescale memory and sequence generation.
- Built a Python simulation and analysis framework from scratch: numerical ODE integration, eigendecomposition pipelines, and Lyapunov spectrum computation by QR reorthonormalization.
- Construct linear networks with many near-critical oscillatory modes and characterize how recurrence time scales with network size, and test whether timed sequences can be decoded by simple ReLU readout neurons.
- Applied Dynamic Mode Decomposition to learn linear dynamics from high-dimensional neural population recordings.

University of New Mexico, Ronald E. McNair Scholar and ASSURE Fellow — 2021 – 2022

- Ran variance-based (Sobol') global sensitivity analysis on the Hodgkin–Huxley neuron model: hundreds of thousands of simulations to estimate first- and total-order indices for eight parameters.
- Presented results at three national McNair research conferences.

Skills and Activities

Programming: Python (NumPy, SciPy, Matplotlib), R, MATLAB, Git, LaTeX, high-performance computing

Statistics and ML: time series forecasting (ARIMA, EMD, seasonal-trend decomposition), sensitivity analysis, k-means clustering, PCA, CNNs

Applied math: numerical ODEs, eigenspectral methods, Lyapunov exponents, Dynamic Mode Decomposition, large-scale simulation

Teaching: Peer Learning Facilitator, Advanced Data Analysis I & II (STAT 427/527, 428/528), UNM, 2021 – 2022

Affiliations: SIAM, American Mathematical Society, Society for Mathematical Biology